Q1.

Satellites are used to gather data about the origin of the Universe.

The Big Bang theory is a theory about the origin of the Universe.

Evidence for the Big Bang theory is provided by red-shift and CMB radiation.

(i) Describe what is meant by red-shift.

(2)

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(ii) Explain how red-shift provides evidence for the Big Bang theory.

(2)

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(iii) The Cosmic Background Explorer (COBE) satellite observed CMB radiation from 1989 to 1993.
State what the 'M' in CMB radiation stands for.

(1)

.....

(iv) State what is meant by 'cosmic background radiation'.

(1)

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(v) Explain how the presence of CMB radiation provides evidence for the Big Bang theory.

(2)

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(Total for question = 8 marks)

Q2.

Figure 7 shows two hydrogen spectra.

One spectrum is taken from a source on Earth.

The other spectrum is taken from a source on a distant galaxy.

The spectral lines in the spectrum from the distant galaxy are shifted towards the red end of the spectrum.

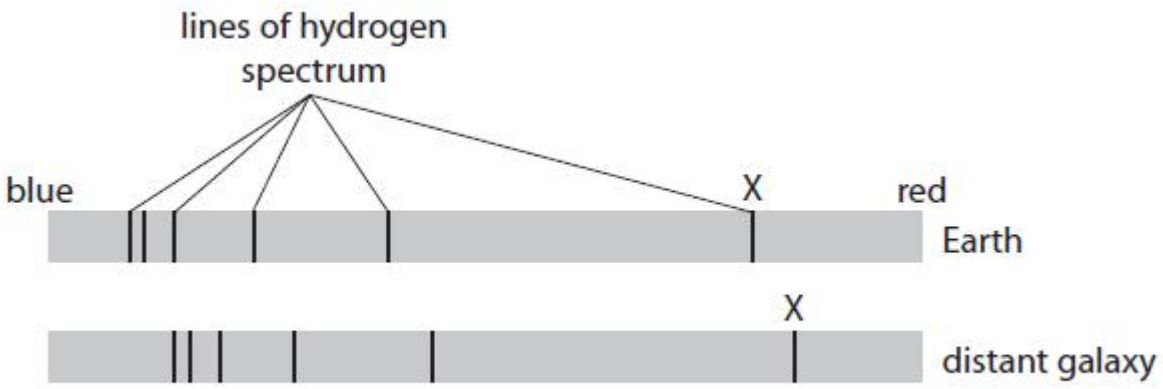

Figure 7

The wavelength of line X on Earth, λ_o = 6.56 x 10^{-7}m

The wavelength of line X from the distant galaxy, λ_g = 6.72 x 10^{-7}m

The red shift (z) is given by the equation

$$z = \frac{(\lambda_g - \lambda_o)}{\lambda_o}$$

(i) Show that the red shift for the light from the distant galaxy is about 0.025

(2)

(ii) The galaxy is moving away from the Earth at velocity, v (recession velocity).

The velocity of light, c = 3.00 x 10^8 m/s
The recession velocity is given by the equation.

$$v = z \times c$$

Calculate the recession velocity of the distant galaxy.

(2)

recession velocity = ... m/s

(iii) The wavelength of the spectral line X measured for a more distant galaxy was 6.92×10^{-7} m.

Explain how this provides evidence that the Universe is expanding.

(2)

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.....

.....

.....

(Total for question = 6 marks)

Q3.

Our Sun is a main sequence star.
It will stay as a main sequence star for millions of years, continuing to radiate energy.

(i) Our Sun was formed from a nebula.

Describe the formation of our Sun from nebula to main sequence.

(2)

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.....

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.....

(ii) Our Sun will go through many stages.

Which of these stages will our Sun go through?
Put a cross (☒) in the box next to your answer.

(1)

☐ **A** a black hole

☐ **B** a neutron star

☐ **C** a supernova

☐ **D** a white dwarf

Q4.

CMB radiation provides evidence that the Universe had a definite beginning.

Use the table in Figure 7 to give a typical value for the wavelength of CMB radiation.

type of radiation	typical wavelength
gamma rays	1.0×10^{-12} m
X-rays	3.0×10^{-11} m
ultraviolet	200 nm
visible	600 nm
infrared	4.0 μm
microwaves	1.0 mm
radio waves	50 m

Figure 7

(2)

wavelength = ...

(Total for question = 2 marks)

Q5.

The table in Figure 16 shows some data about four stars.

star	mass in kg	age in years
Antares	2.6×10^{31}	1.5×10^{7}
Arcturus	2.2×10^{30}	7.1×10^{9}
Proxima Centauri	2.4×10^{29}	4.8×10^{9}
The Sun	2.0×10^{30}	4.6×10^{9}

Figure 16

(i) State which star in the table in Figure 16 is **most likely** to become a Black Hole.

(1)

.....

(ii) Using data from the table in Figure 16, give a reason for your answer to part (i).

(1)

.....

.....

(Total for question = 2 marks)

Q6.

Figure 3 shows a photograph of galaxy D.

This photograph was taken by a student at his home.

(Source: Paul Curtis)

Figure 3

State **two** ways that the student can improve the observational techniques so that the quality of the image is improved.

(2)

1 .....

.....

2 .....

.....

(Total for question = 2 marks)

Q7.

During the evolution of a star, the nebula collapses and becomes a main sequence star.

(i) State what causes the nebula to collapse.

(1)

.....

(ii) Explain why the nebula stops collapsing as it becomes a main sequence star.

(3)

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.....

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.....

(Total for question = 4 marks)

Q8.

Two theories about the Universe are the Steady State Theory and The Big Bang Theory.

(i) The table shows some ideas about the Universe.

Which row of the table applies to the Steady State Theory?
Put a cross (☒) in the box next to your answer.

(1)

	the Universe.....	the Universe.....
☐ A	... is not expanding	... had a beginning
☐ B	... is expanding	... had a beginning
☐ C	... is not expanding	... did not have a beginning
☐ D	... is expanding	... did not have a beginning

(ii) State the name of the discovery that persuaded most scientists to prefer the Big Bang Theory to the Steady State Theory.

(1)

.....

Q9.

Figure 2 shows some lines in the absorption spectra from four different galaxies (A, B, C, and D) and from a laboratory source.

All the spectra are aligned and to the same scale.

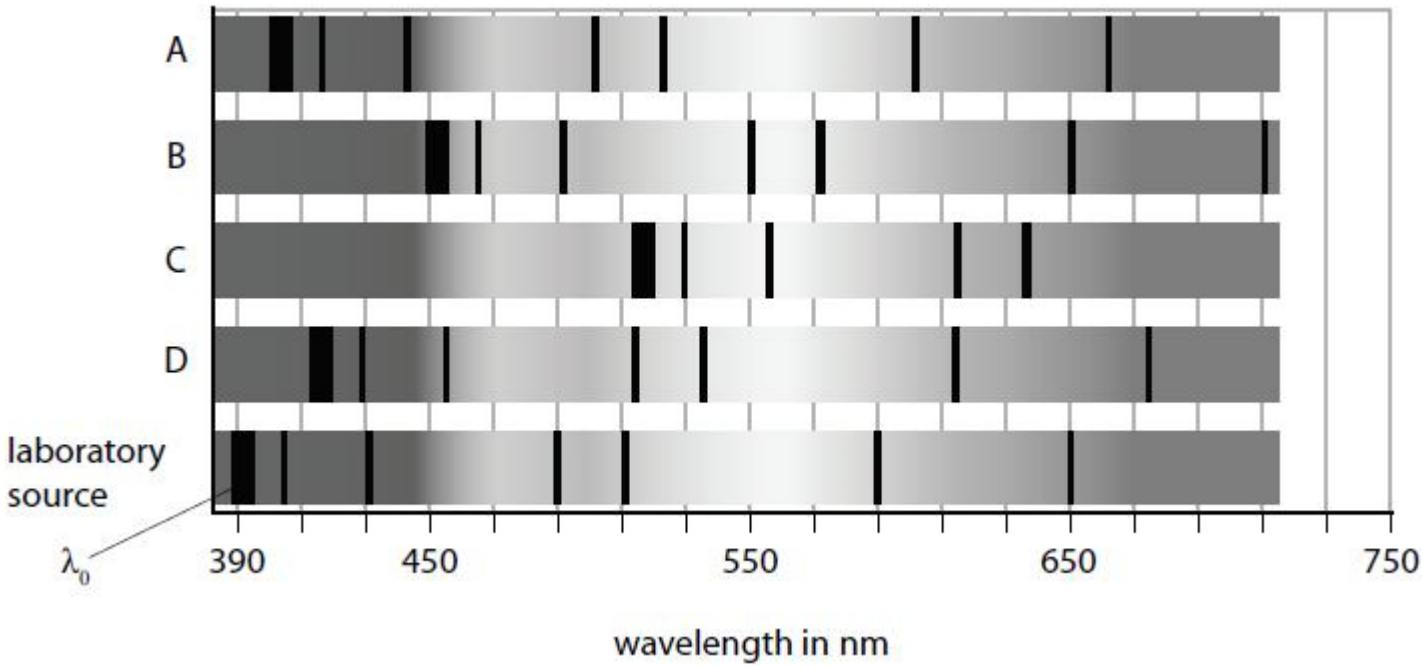

Figure 2

(i) Explain, using Figure 2, which galaxy is furthest away from us.

(3)

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(ii) In Figure 2, the reference wavelength, λ_0, is shown at 390 nm.

Estimate the change in the reference wavelength, $\Delta\lambda$, for the light from galaxy D.

(1)

$\Delta\lambda$ = ... nm

(iii) Calculate the speed, *v*, of galaxy D.

Use the equation

$$v = c\frac{\Delta\lambda}{\lambda_0}$$

[c = speed of light = 3×10^8 m/s]

(2)

v = ... m/s

(Total for question = 6 marks)

Q10.

A star has evolved to become a neutron star.

The mass, M, of the neutron star, of radius R, is given by

$$M = \frac{4 \times \pi \times D \times R^3}{3}$$ where D is a constant

$M = 4 \times 10^{30}$ kg
$D = 6 \times 10^{17}$ kg/m^3
Use the equation to calculate the value for R.

(2)

R = ... m

(Total for question = 2 marks)

Mark Scheme

Q1.

Question Number	Answer	Additional guidance	Mark
(i)	wavelength (of e.m. radiation) increased / frequency decreased (1) as the (star) moves away (from us) (1)	don't penalise planet instead of object stretched/moves to(wards) red end of spectrum spectral lines move to the red end of the spectrum	(2) AO 1 1

Question Number	Answer	Additional guidance	Mark
(ii)	an explanation linking: • big bang has expanding universe (1) • with galaxies moving away (from each other) (1)	from (original) explosion started at a point accept stars moving away (not objects or planets now) the further away they are the greater is their (recessional) speed idea	(2) AO 1 1

Question Number	Answer	Additional guidance	Mark
(iii)	microwave		(1) AO 1 1

Question Number	Answer	Additional guidance	Mark
(iv)	(radiation) that comes from all over the sky / space / the universe	from the big bang / explosion	(1) AO 1 1

Question Number	Answer	Additional guidance	Mark
(v)	an explanation linking: • the Big Bang theory has a beginning / initial explosion (1) • that releases/gives out radiation (1)	 explosion from a point radiation still present	(2) AO 1 1

Q2.

	Answer	Additional guidance	Mark
(i)	substitution (1) $(z=)\frac{6.72\times10^{-7}-6.56\times10^{-7}}{6.56\times10^{-7}}$ **OR** $(z=)\frac{0.16\times10^{-7}}{6.56\times10^{-7}}$ evaluation (1) 0.024	allow $(z=)\frac{6.72-6.56}{6.56}$ **OR** $(z=)\frac{0.16}{6.56}$ do not accept 0.025 on its own accept 0.0243 / 0.02439 / 0.0244 award full marks for the correct answer without working	(2) **AO2**

	Answer	Additional guidance	Mark
(ii)	substitution (1) (v) = 0.024 x 3(.00) x10^8 evaluation (1) 7.2 x 10^6 (m/s)	accept (v) = 0.025 x 3 x10^8 (v) = 0.02 x 3 x10^8 accept answers which round to: 6.0 x10^6 (m/s) 7.5 x10^6 (m/s) 7.3 x10^6 (m/s) 7.2 x10^6 (m/s) award full marks for the correct answer without working	(2) AO2

	Answer	Additional guidance	Mark
(iii)	An explanation linking :- the redshift / z (value) will be larger (1) the galaxy is moving (away) at a higher velocity / recession velocity increased (1)	 bigger (increase in) wavelength bigger difference in wavelength longer wavelength moving (away) faster	(2) AO1

Q3.

Question Number	Answer	Acceptable answers	Mark
(i)	Description including any two of: • gravity (1) • (causes the) nebula to collapse/contract (1) • (causes the) temperature to increase (1)	Pulls {particles/gas} together Forms protostar ke transferred to thermal energy KE/GPE ->thermal GPE -> KE until it was hot enough to start the reaction until fusion starts	(2)

Question Number	Answer	Acceptable answers	Mark
(ii)	D white dwarf		(1)

Q4.

	Answer	Additional guidance	Mark
	1(.0) (1) mm (1)	Allow values between 1.0 and 1.9 allow 1×10^{-3} m or 0.001 m for 2 marks if nothing in answer line, credit answer indicated in table	(2) AO3

Q5.

Question Number	Answer	Additional guidance	Mark
(i)	Antares (1)	accept phonetic spelling	(1) AO3.1

Question Number	Answer	Additional guidance	Mark
(ii)	the greatest/biggest/largest mass (in the table) (1)	(much) larger mass than the Sun / (much) heavier than the Sun allow heaviest ignore "biggest" alone ignore ref to age	(1) AO3.1

Q6.

Question number	Answer	Additional guidance	Mark
	Any **two** from the following improvements: • use wider aperture telescope/camera (1) • better quality objective lens (1) • use longer exposure time while telescope is locked onto star (1) • move telescope to better seeing conditions, e.g. dry desert, higher up a mountain, dark skies (1)	allow improvements from photography, e.g. use longer exposure time use a satellite telescope ignore use pc to adjust the sharpness of the image	(2)

Q7.

	Answer	Additional guidance	Mark
(i)	gravitational attraction / gravitational force (causing collapse) (1)	allow gravity ignore weight ignore gpe ignore gravitational energy	(1) AO1

	Answer	Additional guidance	Mark
(ii)	An explanation linking: (gravity causing) increase in temperature (1) (until hot enough for) fusion (1) (until) balance (between gravity and fusion/thermal) (1)	 allow increase in pressure/density hydrogen to form helium allow nuclear reactions ignore fission allow equilibrium / counteracts	(3) AO1

Q8.

Question Number	Answer	Acceptable answers	Mark
(i)	**D** ..is expanding ... did not have a beginning		(1)

Question Number	Answer	Acceptable answers	Mark
(ii)	Cosmic Microwave Background (Radiation)	[order of words unimportant] CMB(R) reject 'CMB and red shift'	(1)

Q9.

Question number	Answer	Mark
(i)	An explanation that combines identification via a judgement (1 mark) to reach a conclusion via justification/reasoning (2 marks): • galaxy C has the greatest red shift (1) • so this galaxy has the greatest speed (1) • since the galaxy with the greatest speed will be furthest away, then galaxy C is at the furthest distance(1)	(3)

Question number	Answer	Additional guidance	Mark
(ii)	20 (nm)	Allow answers in the range 19 to 25	(1)

Question number	Answer	Additional guidance	Mark
(iii)	Substitution (1) $v = \frac{(3\times10^{8})\times(20\times10^{-9})}{(390\times10^{-9})}$ Answer (1) = 15 400 000 (m/s)	allow ecf from (c)(i) power of 10 error = max 1 accept 15 384 615 (m/s) award full marks for correct numerical answer without working	(2)

Q10.

Question Number	Answer	Additional guidance	Mark
	rearrangement (1) $R^3 = \frac{3M}{4 \times \square \times D}$ evaluation (1) (R =) 1.17×10^4 m	may be seen as substituted values or as the cube-root form or $R^3 = 1.59 \times 10^{12}$ allow numbers that round to 1.2×10^4 (m) award full marks for the correct answer without working	**(2)**